

Lecture 8: Inference for Proportions

STAT 521

Applied Probability and Statistical Inference

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Binomial Data

- A binomial random variable Y is defined as the number of successes observed in n independent trials which resulted in either success or failure.
- The probability of success, denoted by p , is the parameter of the Binomial probability distribution:

$$P(y) = \frac{n!}{y!(n-y)!} p^y (1-p)^{n-y}$$

- The point estimate for the parameter p is simply

$$\hat{p} = \frac{y}{n},$$

which corresponds to the proportion of successes in the sample.

Point estimate for a population proportion

- Each observation can be represented as a Bernoulli random variable:

$$X_i = \begin{cases} 1, & \text{if success} \\ 0, & \text{if failure.} \end{cases}$$

- The sample proportion is therefore the **sample mean** of these Bernoulli trials:

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i.$$

- Because $\hat{p}$ is an average, the *Central Limit Theorem* applies meaning the distribution of $\hat{p}$ is approximately normal.

Sample proportions are also nearly normally distributed

- Using the CLT:

$$\hat{p} \sim N\left(\text{mean} = p, SE = \sqrt{\frac{p(1-p)}{n}}\right)$$

- But of course this is true only under certain conditions...
 - *Independence*
 - Randomization
 - *Nearly Normal*
 - Number of successes ($n \times p$) ≥ 5
 - Number of failures ($n \times (1 - p)$) ≥ 5
- Note p is unknown, we use $\hat{p}$ in the calculation of the standard error and to check assumptions.

Confidence interval for a population proportion

- A $(1 - \alpha)$ **confidence interval** for the population proportion p is:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- The margin of error is:

$$ME = z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- Conditions:

- Independence (randomization)
- Success-failure: $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$

Example: Statistics and the General Population

Two scientists want to know if a certain drug is effective against high blood pressure.

- The first scientist wants to give the drug to 1000 people with high blood pressure and see how many of them experience lower blood pressure levels.
- The second scientist wants to give the drug to 500 people with high blood pressure, and not give the drug to another 500 people with high blood pressure, and see how many in both groups experience lower blood pressure levels.

Which is the better way to test this drug?

- (a) All 1000 get the drug (b) 500 get the drug, 500 don't

Results from the GSS

The General Social Survey asks the same question, below is the distribution of responses from the 2010 survey:

All 1000 get the drug	99
500 get the drug 500 don't	571
Total	670

Parameter and Point Estimates

We would like to estimate the proportion of all Americans who have a good intuition about experimental design, i.e. would answer “500 get the drug 500 don’t”?

What are the parameter of interest and the point estimate?

- *Parameter of interest:* Proportion of *all* Americans who have a good intuition about experimental design.

p (a population proportion)

- *Point estimate:* Proportion of *sampled* Americans who have a good intuition about experimental design.

$$\hat{p} = \frac{y}{n} = \frac{571}{670} = 0.85 \text{ (a sample proportion)}$$

Confidence Interval

The GSS found that 571 out of 670 (85%) of Americans answered the question on experimental design correctly. Using a 95% confidence interval, estimate the proportion of all Americans who have a good intuition about experimental design.

Given: $n = 670$, $\hat{p} = 0.85$. First check assumptions & conditions.

1. **Independence:** The sample is random, therefore we can assume that one respondent's response is independent of another.
2. **Normality:** 571 people answered correctly (successes) and 99 answered incorrectly (failures), both are greater than 5.

We are given that $n = 670$, $\hat{p} = 0.85$, we also just learned that the standard error of the sample proportion is $SE = \sqrt{\frac{p(1-p)}{n}}$. Which of the below is the correct calculation of the 95% confidence interval?

- (a) $0.85 \pm 1.96 \times \sqrt{\frac{0.85 \times 0.15}{670}}$
- (b) $0.85 \pm 1.96 \times \sqrt{\frac{0.85 \times 0.15}{670}} \rightarrow (0.82, 0.88)$
- (c) $0.85 \pm 1.65 \times \sqrt{\frac{0.85 \times 0.15}{670}}$
- (d) $0.85 \pm 1.96 \times \frac{0.85 \times 0.15}{\sqrt{670}}$
- (e) $571 \pm 1.96 \times \sqrt{\frac{571 \times 99}{670}}$

We are 95% confident that the true proportion of Americans who have a good intuition about experimental design is between 82% and 88%.

Choosing a sample size

If the researchers were going to conduct another study on the same survey question how many people should they sample in order to cut the margin of error of a 95% confidence interval down to $\leq 1\%$?

$$CI = \hat{p} \pm ME$$

$$ME = z_{\alpha/2} \times SE = z_{\alpha/2} \sqrt{\frac{p(1-p)}{n}} \approx z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$0.01 \geq 1.96 \times \sqrt{\frac{0.85 \times 0.15}{n}} \rightarrow \text{Use estimate for } \hat{p} \text{ from previous study}$$

$$0.01^2 \geq 1.96^2 \times \frac{0.85 \times 0.15}{n}$$

$$n \geq \frac{1.96^2 \times 0.85 \times 0.15}{0.01^2}$$

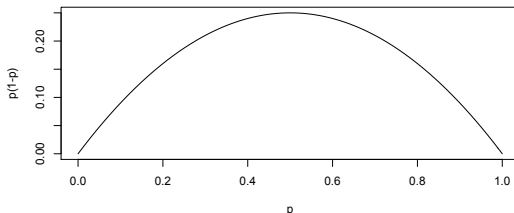
$$n \geq 4898.04 \rightarrow n \text{ should be at least } 4,899$$

What if there isn't a previous study?

What should the researchers do if they are planning to ask a new survey question where they have no idea what the population proportion might be ... use $p = 0.5$

Why?

- if you don't know any better, 50-50 is a good guess
- $p = 0.5$ gives the most conservative estimate because $p(1 - p)$ is largest when $p = 0.5$ which results in the largest possible n



Choosing a sample size, cont.

Previously we just saw that if the researchers want a margin of error less than 1% they will need to sample at least 4,899 people when they expect the population proportion to be near 85%. How does this change when we expect the population proportion to be near 50%?

$$0.01 \geq 1.96 \times \sqrt{\frac{0.5 \times (1 - 0.5)}{n}}$$

$$0.01^2 \geq 1.96^2 \times \frac{0.5 \times 0.5}{n}$$

$$n \geq \frac{1.96^2 \times 0.5 \times 0.5}{0.01^2}$$

$$n \geq 9604 \rightarrow \text{which is almost double the sample size!}$$

Bootstrap confidence interval for a proportion

- Compute the sample proportion:

$$\hat{p} = \frac{y}{n}$$

- Resample (with replacement) B times from the data. For each bootstrap sample b , compute $\hat{p}_b^*$.
- The bootstrap distribution of $\hat{p}^*$ estimates the sampling distribution of $\hat{p}$.
- **Percentile CI:**

$$[\hat{p}_{(\alpha/2)}^*, \hat{p}_{(1-\alpha/2)}^*]$$

- Advantage: no Normal or large-sample assumptions.

Hypotheses Testing

Which of the following are the correct set of hypotheses for testing if more than 80% of Americans have a good intuition about experimental design?

(a) $H_0 : \mu = 0.80$
 $H_A : \mu > 0.80$

(b) $H_0 : p = 0.85$
 $H_A : p > 0.85$

(c) $H_0 : p = 0.80$
 $H_A : p > 0.80$

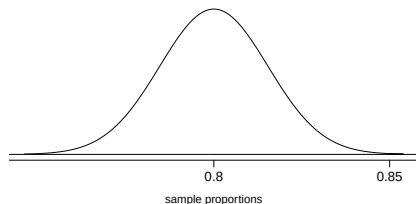
(d) $H_0 : p = 0.80$
 $H_A : p > 0.80$

(e) $H_0 : \hat{p} = 0.80$
 $H_A : \hat{p} > 0.80$

Hypothesis testing for a proportion

$$\hat{p} \sim N\left(\text{mean} = 0.80, SE = \sqrt{\frac{0.80 \times 0.20}{670}} = 0.0154\right)$$

Note: The SE is different, because now we are conducting a hypothesis test assuming H_0 is true, and H_0 says $p = 0.80$.



$$Z = \frac{0.85 - 0.80}{0.0154} = 3.25$$

$$\begin{aligned} p\text{-value} &= P(Z > 3.25) \\ &= 1 - 0.9994 = 0.0006 \end{aligned}$$

Since p-value is less than 0.05 we reject H_0 . The data provide convincing evidence that more than 80% of Americans have a good intuition on experimental design.

Example - Legalizing Marijuana

The 2010 General Social Survey also asked the question; “Do you think the use of marijuana should be made legal, or not?” 48% of the 1,259 respondents said it should be made legal.

- a) Is the number “48%” a sample statistic or a population parameter?
- b) Construct a 95% confidence interval for the proportion of Americans who think marijuana should be made legal $\rightarrow (0.467, 0.494)$.
- c) Interpret this confidence interval in the context of this question.
- d) A critic points out that this 95% confidence interval is only accurate if the statistic follows a normal distribution, or if the normal model is a good approximation. Is this true for these data? Explain.
- e) A news piece on this study's findings states; “Majority of Americans think marijuana should be legalized.” Based on your confidence interval, is this news piece's statement justified?

Example - Legalizing Marijuana

We can also employ a hypothesis test to examine whether a majority of americans support legalizing marijuana.

$$H_0 : p = 0.50$$

$$H_A : p > 0.50$$

$$\hat{p} \sim N\left(\text{mean} = 0.5, SE = \sqrt{\frac{0.50 \times 0.50}{1259}} = 0.014\right)$$

$$Z = \frac{0.48 - 0.5}{0.014} = -1.43$$

$$\begin{aligned} p - \text{value} &= P(Z > -1.43) \\ &= 1 - 0.0764 = 0.9236 \end{aligned}$$

Therefore, we fail to reject the null hypothesis since $p - \text{value} > 0.05$. There is *not* enough evidence that the majority of americans support legalizing marijuana.